Generic Feedback to
January 2022/23 Semester 1 exam
2MVA Multivariable & Vector Analysis 06 25667 / 06 35172

Generally good answers to the challenging exam. Many correct, rigours and well-presented answers!

Q1a is to calculate the rate of steepest ascent and decent of a function at a given point. Most students obtained correct answers for this part, except for calculation mistakes.

Q1b is to find and characterize the stationary points of functions. Most students knew well the leading minor test, but some students did not obtain the stationary points correctly.

Q1c is to change variables for a partial differential equation. It involves using the chain rule due to composite functions and the change of variables. It is a challenging question.

Q3a is to express a double integral as repeated integrals using both Cartesian and polar coordinates, and calculate the integral using anyone of the two coordinates.

Q3ai: Many students obtained correct solution for this part.

$$I = \int_1^2 dx \int_0^{\sqrt{3}x} \frac{dy}{\left(x^2+y^2\right)^{3/2}} .$$

Q3aii: Many students obtained correct solution for this part.

$$I = \int_0^{\pi/3} d\theta \int_{\sec\theta}^{2\sec\theta} \frac{1}{r^3} r\, dr .$$

Q3aiii: Many students obtained correct solution for this part: $I = \sqrt{3}/4$.

Q3b is an integration over a complex three-dimensional domain, involving spatial imagination, using cylindrical and spherical coordinates, and using symmetry properties of a domain. The main technique used here is to draw the cross-section of the domain in the rz (θ = constant) plane, as the domain is axisymmetric to the z-axis. This question requires spatial imagination.

Q3bi: Many students obtained correct solution for this part, $J = \int_0^{2\pi} d\theta \int_0^1 r dr \int_{\sqrt{3}r}^{\sqrt{4-r^2}} e^{\left(r^2+z^2\right)^{3/2}} dz$.

Q3bii: Many students obtained correct solution for this part, $J = \int_0^{2\pi} d\theta \int_0^{\pi/6} d\phi \int_0^2 e^{\rho^3} \rho^2 \sin\phi d\rho$.

Q3biii: $J = \frac{\pi}{3}\left(2-\sqrt{3}\right)\left(e^8-1\right)$.

Q3biv & V: Some students obtained correct solution. One needs to employ and justify the properties for triple integrals for even and odd functions over symmetrical domains.

$K = J$, $L = J/2$.

The common mistakes for Q3 are: set the integral limits wrongly, did not attempt or finish the answer, and did not explain/justify symmetrical properties used.

Where students typically did well: Q1a, Q1b, and Q3a.

General suggestions for improving the results:

Be patient in calculation to avoid mistakes.

Be careful in setting integral limits, since wrong integral limits result in wrong solutions at starting. Perform geometrical analyses and/or draw the figures for integral domains to determine logically the integral limits, not only depending on instinct.

Repeat example questions in the notes, do questions suggested in the problem/exercise sheets and clear any uncertainties with the lecturer. **Practice makes perfect!**